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Review

Emerging and re-emerging themes in co-transcriptional pre-mRNA splicing

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SUMMARY

Proper gene expression requires the collaborative effort of multiple macromolecular machines to produce functional messenger RNA. As RNA polymerase II (RNA Pol II) transcribes DNA, the nascent pre-messenger RNA is heavily modified by other complexes such as 5′ capping enzymes, the spliceosome, the cleavage, and polyadenylation machinery as well as RNA-modifying/editing enzymes. Recent evidence has demonstrated that pre-mRNA splicing and 3′ end cleavage can occur on similar timescales as transcription and significantly cross-regulate. In this review, we discuss recent advances in co-transcriptional processing and how it contributes to gene regulation. We highlight how emerging areas—including coordinated splicing events, physical interactions between the RNA synthesis and modifying machinery, rapid and delayed splicing, and nuclear organization—impact mRNA isoforms. Coordination among RNA-processing choices yields radically different mRNA and protein products, foreshadowing the likely regulatory importance of co-transcriptional RNA folding and co-transcriptional modifications that have yet to be characterized in detail.

RNA SYNTHESIS IS COUPLED TO RNA-PROCESSING STEPS

Eukaryotic mRNAs are transcribed initially by RNA polymerase II (RNA Pol II) in an immature form requiring extensive processing. The first step—5′ end capping—occurs as the ∼15–17 nt long nascent RNA emerges from the exit RNA Pol II exit tunnel, owing to the recruitment of the capping enzymes by the phosphorylated C-terminal domain (CTD) of RNA Pol II and physical interaction with the core polymerase (Figure 1A).[1,2,3] The resulting 7-methyl guanosine cap protects the immature nascent transcript as well as the mature mRNA from 5′ to 3′ exonucleases, binds to nuclear and cytoplasmic cap-binding proteins, assists in targeting RNA for degradation during surveillance steps, and recruits splicing and translation factors.[4] Because it is impossible to remove the RNA Pol II CTD and retain transcription *in vivo*, the proportion of mRNA that would receive the 5′ cap without coupling to transcription is unknown. Presumably, the efficiency would be vanishingly small, as the 5′ end of nascent RNA lacks sequence determinants that might recruit the first of the three required enzymatic activities. The only other way to obtain a 5′ cap is by cytoplasmic re-capping.[5] Thus, most caps are installed in coordination with the transition from transcription initiation to elongation. Upon RNA Pol II reaching the end of the gene, the pre-mRNA must be endonucleolytically cleaved, determining the templated 3′ end of the mature isoform and usually triggering polyadenylation.[6] The selection of alternative last exons or alternative poly(A) (APA) sites can lead to differential regulation, shaping the fate of the resulting transcript with direct implications on coding potential, localization, and transcript stability.[7] Cleavage at a polyadenylation site initiates termination of transcription by the attached polymerase, making this step co-transcriptional by definition. Accordingly, structural and crosslinking studies have placed the cleavage and polyadenylation machinery in close proximity to RNA Pol II.[8] In summary, 5′ end capping and 3′ end cleavage steps are entirely co-transcriptional, determine the two ends of the mature mRNA, and effectively define the annotated transcription start and end sites in genome annotations.

By comparison to capping and 3′ end cleavage, pre-mRNA splicing carried out by the spliceosome makes the most drastic changes in the body of the (pre-)mRNA.[10] The removal of intervening non-coding regions (introns) that are interspersed among exons dramatically remodels the pre-mRNA, which theoretically matches the sequence of its parent gene, into an RNA with a radically different and much shorter sequence. While gene architecture differs among organisms, introns can be plentiful in eukaryotic genomes with humans typically having 8 introns per gene.[11] Furthermore, introns make up the majority of pre-messenger RNA (pre-mRNA) transcripts because the median length for introns is 10 times greater than for exons.[12,13] Variability in intron removal, producing alternatively spliced isoforms or intron-retained mRNAs, occurs in most multicellular organisms and contributes significantly to RNA localization, proteome diversification, and gene regulation.[14–18]

Global analyses have revealed that ∼75% of introns are removed during transcription, while ∼25% of introns are removed after transcription termination, i.e., post-transcriptionally.[19–25] Accordingly, the spliceosome is able to operate on the timescale of transcription in a variety eukaryotes ranging from yeast to humans.[19,20,22,26–29] Close proximity during synthesis and processing can lead to spatial and kinetic coupling that afford the opportunity for cross-regulation and cooperativity among the synthesis and RNA-modifying machinery. For

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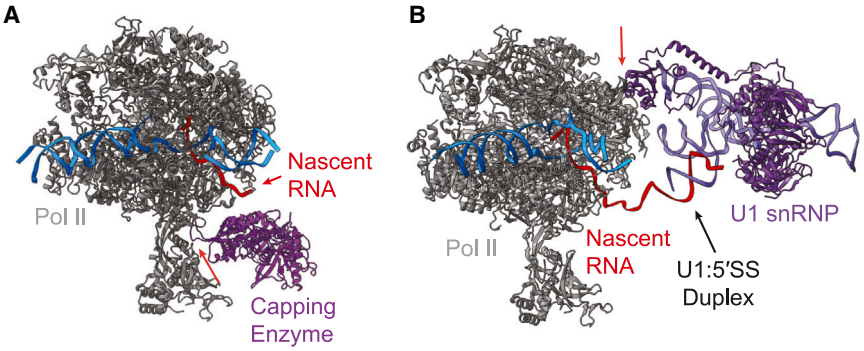

Figure 1. RNA polymerase II contacts nascent RNA-modifying complexes

(A) Cryo-EM reconstruction of RNA Pol II and the 5′ capping enzyme shows physical interaction at the RPB4/RPB7 stalk and RPB1 (orange arrow) during RNA capping (PDB: 8W8E).[1] Note that the NELF complex observed in the reconstruction is omitted for clarity.

(B) Cryo-EM structure of transcribing RNA Pol II (gray) interacting with human U1 snRNP (purple and orange). U1 contacts RNA Pol II directly via an interaction between U1-70K and a joint interface formed by Rpb2 and Rpb12 (contact indicated with orange arrow) (PDB: 7b0y).[9]

instance, some splicing factors interact—either directly or indirectly—with the transcription machinery, including the RNA polymerase core and/or the CTD (Figure 1B).[9,22,30,31] Inclusion of an intron in transgenes positively correlates with increased transcriptional activity, whereas altering the speed of RNA Pol II transcription leads to changes in splicing outcomes.[32–34] The past three decades of work have sought to understand co- and post-transcriptional processing using a variety of *in vivo* and *in vitro* approaches to generate a holistic understanding of the molecular events required for efficient gene expression. This review focuses on recent advancements in the field's understanding of the molecular mechanisms that couple co-transcriptional pre-mRNA processing events and their consequences, focusing primarily on pre-mRNA splicing.

ASSEMBLY OF THE SPLICEOSOME

Decades of genetic, biochemical, and biophysical studies have established the molecular underpinnings of pre-mRNA splicing, and recent advances in cryo-electron microscopy (EM) have revealed the molecular architecture of the spliceosome throughout the splicing reaction.[35] The spliceosome is a multi-megadalton single-turnover enzyme consisting of five U-rich small nuclear ribonucleoprotein complexes (U1, U2, U4, U5, and U6) and more than a hundred protein-only factors. The spliceosome assembles anew on every intron, meaning most human genes require the assembly of multiple spliceosomes to generate a mature transcript, while only one RNA Pol II holoenzyme is needed for transcription. The spliceosome identifies the intron for removal through the recognition of *cis*-element splice site sequences (i.e., the 5′ splice site [5'ss], the 3' splice site [3′ss], and the branch site [BS]) located at or near intron-exon boundaries (Figure 2A). The spliceosome shares a common mechanism with and is evolutionarily related to the group II self-splicing intron, and intronic RNA (i.e., the splice sites) forms part of the spliceosomal active site.[36] Splice sites are repeatedly recognized throughout assembly of the catalytically active spliceosome to ensure accurate intron borders and maintain the integrity of the encoded message. Splicing must occur with single nucleotide precision to avoid introducing a frameshifting event in the encoded protein. Mutations in the splice sites that alter or weaken base pairing potential with spliceosomal small nuclear RNAs (snRNAs) can cause defective spliceosome assembly, altered splicing outcomes, and ultimately disease. Approximately 14% of all disease-associated point mutations occur in splice site sequences and alter splicing outcomes; that number increases to nearly 50% when accounting for splicing enhancer or silencer elements in the RNA that serve to promote or repress spliceosome assembly, respectively.[37–39]

The spliceosome is compositionally and conformationally dynamic during assembly and catalysis. The field's structural understanding of spliceosomal composition, architecture, and mechanism has been expertly reviewed elsewhere.[10] The spliceosome assembles in an ordered manner from preassembled components on the pre-mRNA.[40,41] A hallmark of early spliceosome assembly is the long-range interaction between splicing factors that are hypothesized promote spliceosome formation. U1 snRNA base pairing with the intron 5′ss is often the first step in spliceosome assembly (Figure 2B). During this process, U1 is stabilized by cross-exon interactions with cap-binding complex bound at the 5′ end of the pre-mRNA and cross-intron interactions with splicing factors (BBP/MUD2 in yeast or SF1/U2AF in humans) that recognize the BS and the polypyrimidine tract characteristic of metazoans located near the 3′ end of the intron, resulting in the spliceosomal E complex.[42–47] E complex is followed by A complex formation in which recruitment of the U2 small nuclear ribonucleoprotein particle (snRNP) to the BS/3′ss region results in release of BBP/MUD2 to allow for U2 snRNA:BS base pairing. Again, cross-intron interactions occur between the U1 and U2 snRNPs to help stabilize the early complex.[48–51] The addition of U4/U6.U5 tri-snRNP to the A complex results in the formation of the pre-B complex, which is converted B complex upon unwinding of the U1:5′ss duplex, ejection of U1, and formation of the U6:5′ss duplex. Subsequent ATP-dependent rearrangements lead to the formation of the spliceosome active site that catalyzes two transesterification reactions (5′ss cleavage and exon ligation) to simultaneously excise the intron and ligate the exons.[52,53]

For multi-intron genes, spliceosome assembly can also proceed via an exon-defined pathway (as opposed to the intron-definition pathway described above).[54,55] In this process, U2 binds the BS located upstream of the defined exon (e.g., exon 2), and U1 binds at the 5′ss located in intron 2. Serine and arginine-rich (SR) proteins stabilize U1 and U2 in these positions by binding along the internal exon.[56] Importantly, cross-exon complexes can recruit U4/U6.U5 tri-snRNP to form pre-B complexes that proceed to catalysis.[56,57] Exon definition stimulates processing

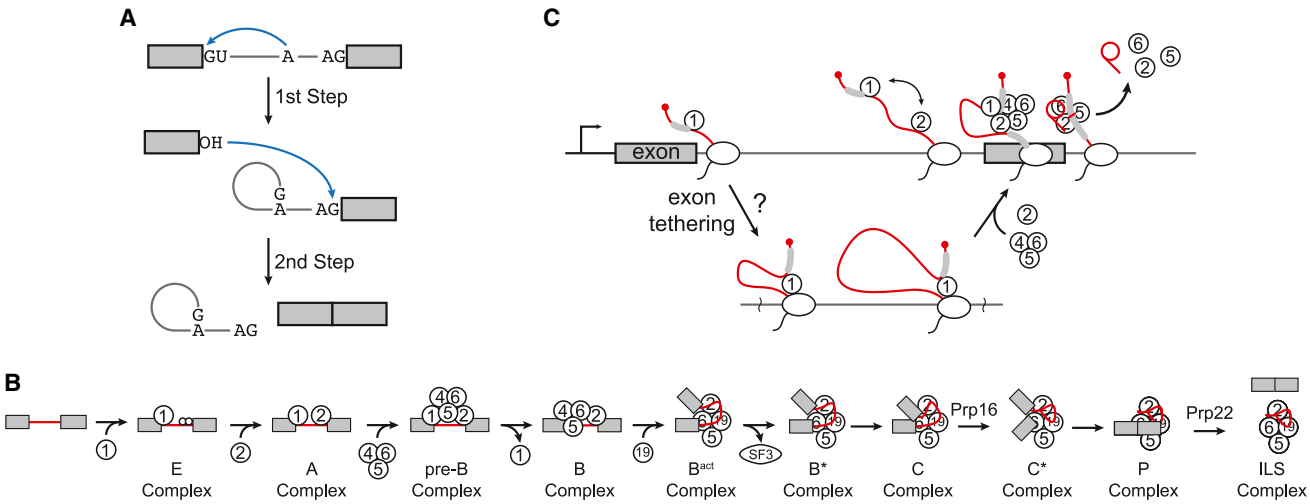

Figure 2. Assembly and catalysis by the spliceosome

(A) Schematic showing the chemical steps of splicing and how the splice sites are used during the catalytic steps.

(B) Diagram showing assembly, activation, and catalysis of the spliceosome. Assembly details are reviewed elsewhere.[10]

(C) Cartoon displaying co-transcriptional spliceosome assembly. For simplicity, many of the steps shown in (B) are abbreviated. U1 snRNP binds the 5′ ss located near the exon 1-intron boundary. Transcription of the downstream intron allows for recruitment of the U2 snRNP to the nascent followed by addition of the U4/U6.U5 tri-snRNP. Catalysis occurs during transcription of the second exon, resulting in the formation of the post-catalytic spliceosome containing the ligated exons and the excised intron lariat. Disassembly of the spliceosome allows for recycling of spliceosomal components for the next round of splicing. U1 or other splicing factors may tether exon 1 to RNA Pol II during transcription of the intron, resulting in a looped-out RNA strand (bottom).

of upstream introns, likely as the result of U1 occupancy at both 5′ss flanking the 3′ss synergistically stimulating spliceosome assembly.[58,59] However, it is important to note that the upstream intron can be committed to splicing without splicing catalysis having occurred; thus, the exon definition does not enforce a strict order of intron removal even though it has already predetermined splicing outcome.[60] Exon definition is thought to occur more commonly in organisms such as mammals that have long introns and relatively short exons.[61] However, recent work has demonstrated that the removal of long introns can also proceed via an intron-defined pathway, leaving some question how significantly the standard concept of exon definition contributes to co-transcriptional splicing *in vivo*.[62] For example, exon sequences can contribute independently to 3′ss recognition without the involvement of the downstream 5′ss.[63] Exon definition may contribute significantly to post-transcriptional splicing, wherein spliceosome assembly on retained introns helps to recruit splicing factors to other introns within the pre-mRNA.

Spliceosome assembly and the order of splice site appearance proceed in an analogous manner, meaning that 5′ss synthesis precedes BS or 3′ss synthesis just as U1 binding precedes binding of other splicing factors like U2 (Figure 2C). Interestingly, in an *in vitro* single-molecule study of spliceosome assembly on post-transcriptional pre-RNAs, it was found that U2 recruitment can precede U1 in a substrate-dependent manner and still result in the formation of active spliceosomes.[64] Nevertheless, co-transcriptional assembly may avoid disadvantageous alternative spliceosome assembly pathways *in vivo*, particularly for higher organisms that have much longer introns and a greater potential for competing cryptic splice sites. Very long introns have been proposed to be removed in a recursive splicing pathway, in which the intron is excised piecewise during multiple rounds of splicing until joining the exons.[65,66] Recursive splicing has been observed in *Drosophila melanogaster*, and a recent study suggests that it frequently occurs in the removal of long human introns.[28,65,67] Using imaging-based approaches to monitor intron removal, Wan et al.[28] were able to show that recursive splicing occurs co-transcriptionally because intron removal often precedes transcription of the entire intron. Whether co-transcriptional spliceosome assembly at recursive sites differs from canonical spliceosome assembly is an interesting question that requires further investigation. One similarity appears to be the involvement of the U2AF2 binding to intronic polypyrimidine tracts near recursively used BS sequences.

CHALLENGES AND ADVANTAGES OF CO-TRANSCRIPTIONAL SPLICEOSOME ASSEMBLY

How does the spliceosome assemble on a nascent RNA when the conventional model requires both ends of the intron for stable snRNP recruitment? Previous work has proposed a model of U1 bridging to RNA Pol II via auxiliary splicing factors, such as FUS.[68] U1 binding early during transcription of the intron would precede the synthesis of the binding sites of factors known to stabilize U1 (e.g., BBP or SF1) (Figure 2C). An interaction with RNA Pol II might replace this function and retain U1 at the site of transcription until synthesis of the intron 3′ end. Interestingly, a recent cryo-EM structure of an *in vitro* assembled complex of human U1 and RNA Pol II identified an apparent physical interaction between U1-70K (Snp1 in yeast) and the Rpb2 and Rpb12 subunits of RNA Pol II.[9] The U1 snRNP contacts near the RNA exit channel, allowing for protein:protein interactions directly with RNA Pol II and simultaneous base pairing with the nascent pre-mRNA 5′ss (Figure 1C). This hypothesis is consistent with the strength of

this interaction as judged by the formation of the U1:RNA Pol II complex even in the absence of base pairing potential with the nascent RNA. This structure has reinvigorated an exon tethering model previously proposed from work using an internal ribozyme in the pre-mRNA to cleave the nascent transcript.[69,70] In this revised model, the exon is tethered to the elongation complex via U1, which rides along the gene as the growing pre-mRNA is looped out until synthesis of the 3′ss.[71] Interestingly, a requirement for an interaction with RNA Pol II to retain stable U1 binding would permit only a single U1 to associate with the nascent transcript at a given time, and this may result in the ordered removal of introns in a 5′ to 3′ manner as has been observed for multi-intron genes.[23,62] Furthermore, screening thousands of sequence combinations found within coding genes revealed that a 5′ss consensus sequence in a nascent RNA stimulates RNA Pol II transcription, possibly due to the recruitment and interaction with U1, suggesting positive feedback between the assembling spliceosome and RNA Pol II.[72,73] This modulation of transcription may exist for other splicing factors as well; disease-associated mutations in the U2 snRNP component SF3b1 reduce elongation rate genome-wide by stalling early steps in spliceosome assembly, possibly physically holding back the elongating RNA Pol II.[74] Interestingly, CDK11—a kinase that modulates transcription—also phosphorylates SF3b1, and this phosphorylated state is characteristic of the active spliceosome.[75,76] Co-transcriptional spliceosome assembly may act as a mechanism to promote productive transcription in the synthesis of long introns by reducing back-splicing, which gives rise to circular RNAs or RNA Pol II backtracking or pausing.

Given rapid intra-molecular base pairing kinetics, RNA that is transcribed and extruded from RNA Pol II will rapidly fold into structures that can be stable enough to occlude splice sites and limit splicing efficiency.[77] Co-transcriptional spliceosome assembly can obviate the need to actively unwind the newly transcribed RNA to allow splicing factors to search for their cognate sequences before extensive folding, but this process requires tight coordination between transcription, splice site recognition, and spliceosome assembly. Transcription speed is non-uniform throughout the gene body, with RNA Pol II transcribing more quickly over introns than over exons, and this is proposed to play in RNA folding and processing.[78,79] Single-point mutations in RNA Pol II that alter transcription speed lead to changes in splicing outcomes in many organisms and even for simple yeast genes.[19,80–82] Changes in transcription rate have also been proposed to play a role in changes in RNA splicing patterns observed in a variety of cancer types.[83] Therefore, there is clear optimization between transcription speed and RNA processing, allowing for product gene expression. Despite its importance, methods for studying how RNA folds during transcription *in vivo* are currently not available. We anticipate exciting discoveries in the area of co-transcriptional structure formation that will be able to reveal how folding is coordinated with transcription speed and numerous RNA-processing steps.

COORDINATED INTRON REMOVAL

As methods became available to ask how individual transcripts behave within the population of nascent pre-mRNAs, it became possible to ask if introns are removed in the order of their synthesis and whether the removal of introns in the same transcript occurs with similar efficiency. The Fairbrother lab published the first study revealing the coordinated removal of introns by analyzing multiple splice junctions from paired-end sequencing of total RNA.[84] Splicing coordination might be the result of the speed of intron removal in each transcript. For example, transcripts in which all introns are removed rapidly will appear coordinated and "all spliced," whereas they will appear "all unspliced" if intron removal is uniformly slow over the whole transcript. In 2011, native elongating transcript sequencing (NET-seq) established the methodology in budding yeast for marking the catalytic site of RNA Pol II by ligating a linker to the 3′-OH of purified nascent RNA; hydrolyzed or nucleolytically cleaved RNAs should not receive the linker because these RNAs usually contain a cyclic 3′ end phosphate that precludes ligation.[85] Whether starting from an RNA Pol II pull-down or a chromatin preparation, linker ligation enables the association of the position of RNA Pol II with upstream processing events, such as splicing, and measuring the distance of RNA Pol II from a splice junction serves as a proxy for splicing speed. Early studies in *Schizosaccharomyces pombe* (*S. pombe*), mouse, and human cells documented coordination of splicing in individual transcripts, revealing that some genes tend toward less efficient splicing of all of their introns while other genes have very efficient removal of all introns.[20,23,29,86] The fate of the efficiently versus inefficiently spliced transcripts is a topic of active research. One possibility is that inefficiently spliced transcripts are degraded by the nuclear exosome, as degradation is in competition with splicing in the yeasts.[23] In mammalian cells, the fate of inefficiently spliced transcripts is difficult to pin down because the action of the nuclear exosome on these transcripts has yet to be characterized and it is difficult to derive (pre-)mRNA half-lives from metabolic labeling approaches from mixed spliced and unspliced transcripts. Remarkably, mammalian erythroid-specific genes such as β-globin display high proportions of transcripts (∼50%) that are completely unspliced and not cleaved at their 3′ ends, while the remaining ∼50% are efficiently spliced and cleaved.[20] This finding resembles the short-read study of Pandya-Jones et al. working with nascent RNA isolated via chromatin fractionation of LPS-treated macrophages, in which unspliced transcripts were shown to exhibit delayed 3′ end cleavage compared with the rapid cleavage of rapidly spliced genes.[87] These phenomena raise the question of how splicing and 3′ end cleavage are coordinated mechanistically at the single transcript level.

APPROACHES TO MONITOR CO-TRANSCRIPTIONAL SPLICING

Some of the earliest evidence to support a model of co-transcriptional processing was the transcription-dependent recruitment of the U1 and U2 snRNPs to fruit fly polytene chromosomes, using monoclonal antibodies that target the Sm antigen and immunofluorescence microscopy.[88] This was supported by the direct observation of ribonucleoprotein (RNP) and splicing factor recruitment to *Drosophila* nascent RNA, using the Miller chromatin spreading technique for ultrastructural

analysis.[89,90] Importantly, this approach identified more rapid spliceosome assembly compared with *in vitro* splicing systems as well as observed exon ligation prior to 3′ end formation, suggesting a significant rate enhancement for *in vivo* splicing.[89] Later experiments in yeast relied on the close association of splicing factors with chromatin to permit crosslinking and chromatin immunoprecipitation (ChIP) to identify the regions along a gene in which splicing factors are recruited. These "splicing factor ChIP" experiments demonstrated snRNP association *in vivo* is consistent with the *in vitro* assembly pathway of the spliceosome, beginning with U1 peaking nearest the 5′ end of the gene downstream of the exon1-intron boundary, followed by U2 and then U4/U6.U5 peaking in the second exon.[91–93] Nevertheless, the low resolution of ChIP data and slow release of post-catalytic spliceosome led to the erroneous conclusion by some studies that while yeast spliceosome assembly occurs co-transcriptionally, exon ligation occurs after poly(A) cleavage.[91,94] Subsequent work using high-density tiling arrays and high-throughput short-read sequencing approaches found that the majority of intron-containing yeast genes are co-transcriptionally spliced.[19,79]

Much of the work over the last decade has focused on understanding the co-transcriptional splicing reaction *in vivo*, using a combination of diverse approaches that involve metabolic labeling, biochemical fractionation, live cell imaging, and next-generation RNA sequencing (reviewed in Shine et al.).[95] Specific emphasis has focused on understanding: (1) how rapidly do spliceosomes assemble on the nascent RNA? (2) When does exon ligation occur? (3) What fraction of transcripts are co-transcriptionally spliced in different species and contexts? (4) What feedback and coupling mechanisms exist between complexes that regulate gene expression? Most modern methods that yield co-transcriptional splicing efficiencies rely on either metabolic labeling with 4-thiouridine to calculate intron half-lives or simultaneously detecting splice junctions and RNA Pol II position in single sequencing reads.[95,96] A surprising outcome of these studies has been the observation of distinct classes of RNA-processing events occurring on very different timescales: rapid and delayed co-transcriptional splicing and post-transcriptional splicing. The boundary between delayed co-transcriptional splicing and post-transcriptional splicing is currently unclear.

RAPID AND DELAYED SPLICING

Rapid splicing is broadly defined as the ligation of exons when the RNA Pol II active site is within a few hundred nucleotides of the newly formed splice junction. Support for the rapid onset of splicing after synthesis of the 3′ss largely comes from RNA sequencing-based approaches that capture nascent transcript ends and spliced status within a single read (specific methods are reviewed in Shine et al.[95] and Merens et al.[96]). This phenomenon was first identified in budding yeast genes using single-molecule intron tracking (SMIT)—a short-read Illumina sequencing-based approach that quantifies fraction of transcripts spliced relative to RNA Pol II position.[19] SMIT identified that the fraction of transcripts spliced from most genes reached a plateau before RNA Pol II was more than 130 nucleotides downstream of the 3′ss. Rapid splicing has also been observed in other organisms using alternative short-read methods such as mammalian NET-seq (mNET-seq) that rely on co-immunopurification of fragments of RNA protected by the RNA Pol II footprint. Spliced junctions can be observed in short RNAs that co-purify with RNA Pol II from Hela cells using antibodies that target phosphorylated Ser5 of the RNA Pol II CTD, possibly because the spliceosome helps to protect them during nuclease treatment.[22] A related method, dNET-seq, was adapted to *Drosophila* embryos and showed instances of rapid splicing.[97] Third-generation long-read sequencing approaches have identified rapid splicing in the distantly related fission yeast species, *S. pombe*.[23] Rapid splicing was also found to be common in mouse and human cell lines.[20,21,29] In particular, >75% of splice junctions were found to be within 300 nucleotides of the read 3′ end for mouse erythroleukemia cells. Together, these studies establish that rapid splicing occurs in a variety of diverse species.

In contrast to rapid splicing, most works that identified delayed splicing largely rely on metabolic labeling using 4-thiouridine to isolate newly transcribed RNA for analysis. For example, the half-lives of both chemical steps of splicing have been estimated using 4-thiouracil treatment in budding yeast, resulting in an estimate for entire splicing reaction of ∼2 min.[98] This value likely represents an overestimate of the time required for splicing, as it assumes all introns synthesized are on pathway to productive splicing. Yeast transcripts with retained introns are known to be withheld for some time in the nucleus and hyperadenylated and/or degraded by the exosome.[99,100] Furthermore, transcription elongation in yeast is estimated to be ∼1.5 kb/min, meaning that RNA Pol II would be significantly farther away upon formation of the spliced junction compared with any of the sequencing-based approaches.[101] Other methods relying on nanopore-based long-read sequencing in human and *Drosophila* cells (S2) also observed spliced junctions forming when RNA Pol II was multiple kilobases away (4 and 2 kb, respectively).[29] Concerns associated with metabolic labeling are that incorporation of uridine analogs can impede splicing, potentially causing the process to appear slower,[102] and that biases toward unspliced are introduced owing to the U-richness of introns. Fortunately, rapid and delayed splicing can be detected in methods that do not rely on metabolic labeling, such as co-transcriptional lariat sequencing (CoLa-seq).[62] The validation of both rapid and delayed splicing kinetics from human tissue culture cells by CoLa-seq creates a unifying conclusion that both phenomena co-exist.

We have much to learn about the rules that govern rapid versus delayed splicing, and nascent RNA recognition by *trans*-acting factors is likely a large part of this story, exemplified by the importance of U2AF binding during recursive splicing (see above). Indeed, delayed splicing may result from the post-transcriptional recruitment of splicing factors or from the delayed onset of catalysis following the rapid co-transcriptional recruitment of spliceosome components. Additional features, such as co-transcriptional RNA modification and folding, are not well understood at a mechanistic level.[95,103–105] However, several examples of disease-causing mutations in folded splicing regulatory elements as well as a functional role for pseudouridine in tuning U2AF recognition are "tip-of-the-iceberg" indicators that *cis* regulation represents the next big wave of insight into

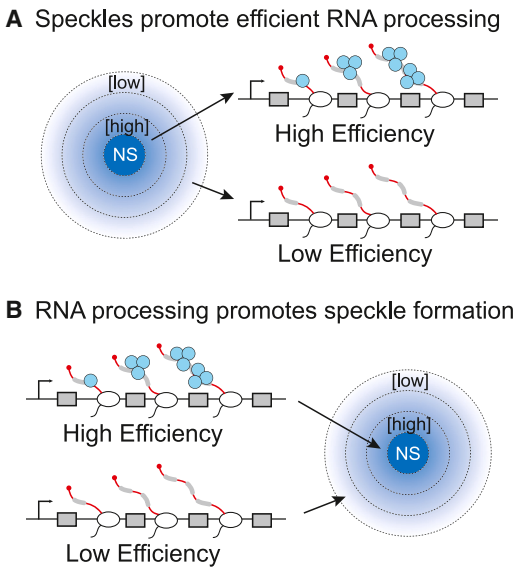

Figure 3. Proposed mechanism for speckles in the enhancement of gene expression

(A) High-efficiency genes are located near a nuclear speckle (NS), thereby increasing transcription and splicing. Genes farther from the NS are less efficiently transcribed and processed as a result of the lower concentration of transcription and splicing factors.

(B) High-efficiency genes recruit high local concentrations of transcription and splicing factors, resulting in the formation of nuclear speckles nearby. The arrow direction depicts the causal factor in the relationship.

splicing regulation.[106,107] An apt example of the potential impact of RNA folding on splicing kinetics was revealed when polypyrimidine tracts were experimentally sequestered in stable stem loops that prevented co-transcriptional 3′ ss recognition and forced post-transcriptional splicing of transcripts.[108] This blockade of U2AF-binding sites thereby disabled exon definition, illustrating the apparent flexibility of the gene expression system that can encompass both rapid as well as delayed intron removal. Additionally, the post-transcriptional splicing observed in this work confirms that co-transcriptional splicing is not necessarily a requirement of gene expression. This principal was initially demonstrated by a cleverly engineered recursive splicing assay in budding yeast.[109] The relative importance of the decision to splice co- or post-transcriptionally to gene expression will be a topic of future studies as more fine-grained platforms for detecting and quantifying the timing of splicing relative to transcription become available.

The best quantifications of co-transcriptional splicing occurrence come from rich datasets, as in SMIT, that enable fractional determinations like "% spliced" that can be obtained with uniform read density over the range of lengths necessary to encompass unspliced and spliced transcripts.[19] In studies that have lower read coverage or length limitations that preclude quantification of the unspliced fraction, it is common to use a cumulative plot of 3′ end distances; at minimum, the distribution of the observed splicing occurrences with respect to RNA Pol II position reveals how quickly introns *can* be spliced. A current limitation of long-read sequencing-based approaches to monitor splicing occurs during library preparation that generally favors the selection of shorter products for two reasons: (1) reverse transcription is not generally processive beyond ~6 kb, and (2) PCR amplification steps tend to enrich for shorter transcripts. This size bias is expected to favor spliced reads, especially for genes with very long (e.g., >6,000 nt) introns. Thus, new approaches will be required to understand this process across vastly different gene architectures. Interestingly, quantification of co-transcriptional splicing occurrence in *S. pombe* is limited by the shortness of introns in this species.[23] The field hopes advances will come with improvements to direct RNA-seq, a long-read sequencing method that in principle does not require reverse transcription of the entire transcript for sequencing to proceed through nanopores.[110] Direct RNA-seq will eventually allow us to determine the structure and processing histories of very long nascent RNAs, including their RNA modifications that can be base-called within the nanopore.

Although some authors suggest that the different papers published on rapid versus delayed splicing are in conflict, there is no doubt that there are many introns that are removed when RNA Pol II is a short distance downstream of introns (i.e., rapid onset). There is also no doubt that many introns are either retained, delayed, or spliced post-transcriptionally. These phenotypes can be observed by multiple sequencing strategies applied to nascent RNA preparations, including short read, long read, mNET-seq, POINT-seq, and PRO-seq (see the preceding paragraphs). What remains unclear is how significantly either rapid or delayed splicing contributes to gene expression.

CO- AND POST-TRANSCRIPTIONAL PROCESSING IN THE 3D NUCLEUS

All nuclear RNA-processing events take place among the nuclear milieu of membraneless organelles and genomic DNA that occupy and make up the nuclear space. There has been substantial effort over the last three decades to understand RNA processing not just during transcription but in the greater context of genome organization and nuclear structure. Nuclear speckles are a membraneless organelles implicated in the co- and post-transcriptional processing of pre-mRNAs (recently reviewed in elsewhere).[111–113] Speckles are enriched in post-transcriptional poly(A)$^+$ RNA and proteins involved in RNA processing, including RNA-binding proteins, splicing factors, and export factors. Multiple studies have indicated that genes close to nuclear speckles are more transcriptionally active and transcripts are more efficiently processed.[114–116] However, it remains unclear whether speckle proximity is the cause or effect of efficient RNA processing. In other words, do speckles assemble on their own, and transcription units find their relative positions with respect to speckles to adjust splicing efficiencies (Figure 3A)? Or does pre-mRNA processing recruit the splicing factors to sequences tethered to chromatin, thereby creating speckle domains near genes with efficient processing (Figure 3B)?

Establishing causality is difficult, but several observations suggest that transcription and splicing play a role in the formation of biomolecular condensates. First, the observation that transcription initiation-related condensates are formed at active genes when the hyperphosphorylated CTD interacts with mediator;

these transition to splicing condensates rich in phosphorylated SR proteins when the CTD is phosphorylated in concert with elongation.[117] Second, phosphorylation of SF3b1 marks active spliceosomes that concentrate in nuclear speckles, providing evidence that pre-mRNAs and spliceosomal components depart from speckles after splicing.[75] Third, imaging of large introns *in situ* shows how extensively nascent RNA can pervade the interchromatin space, supporting the idea that splicing factor recruitment to pre-mRNA could be a major factor in speckle assembly.[118] Furthermore, speckles likely play a role in post-transcriptional splicing, as inhibition of splicing results in targeting of post-transcriptional poly(A)$^+$ RNA to the speckle.[119,120] Taken together, whether speckles are a cause or consequence of efficient processing remains an active area of research.

CONCLUSIONS AND FUTURE PERSPECTIVES

The evolutionary diversity of gene architectures and the flexibility with which the spliceosome can assemble to support alternative splicing are remarkable.[121] Here we have discussed a flexibility in the dynamics of splice site recognition with respect to the order of synthesis of the elements, such as splice sites, exons, and introns. How do genes and/or individual introns "decide" to deploy rapid and delayed co-transcriptional splicing? How are intron splicing events coordinated with each other and with 3′ end cleavage? These broad questions are ripe for mechanistic biochemistry and molecular biology studies, and insights should come within the next few years. A previous study suggested that spliceosomal components retained on the spliced RNA can enhance splicing elsewhere in the transcript, inviting further mechanistic studies that would be facilitated by technological developments.[122] We have mentioned how crucial long-read sequencing is for understanding the coordination of co-transcriptional RNA processing and endorsed direct RNA-seq as a way to obtain longer nascent RNA sequence information. In addition, direct RNA-seq will enable characterization of RNA modifications in nascent transcripts, telling us when during transcription many chemical modifications are installed.[123] Alternatively, recoding chemistries may make the characterization of some nascent RNA modifications detectable in cDNA copies. Previous studies have shown that RNA modifications influence alternative splicing by changing the affinities of splicing factors for target sites, so we believe there is much to learn through these approaches.[106]

It is interesting to consider how different patterns of co-transcriptional coordination are selected by evolution. One must assume these regulatory pathways are selected for because they set the levels of transcripts expressed by cells (like a transcription factor) or produce new proteins through alternative splicing (like the evolution of a new gene). A previous assay that attempted to recapitulate co-transcriptional splicing *in vitro* showed that mRNAs were more stable when produced under co-transcriptional conditions.[124] This has not been tested *in vivo*. A related central question is what fraction of alternative mRNA isoforms generated by splicing is functional? Altered function is clearly selectable by evolution. Human genes have an average of ~3 alternative isoforms, and these numbers are expanding as long-read sequencing of mRNA is being applied to transcriptomes across organisms and tissues to catalog all alternative mRNA isoforms.[125,126] Here progress is being made by the Long-read RNA-Seq Genome Annotation Assessment Project Consortium in developing the analysis tools needed to classify alternative isoforms.[127] Similarly, specific tools must be adapted or developed for long-read sequencing of nascent RNA, which differs from mRNA by having 3′ ends distributed throughout gene bodies. Very likely, these developments will help to pinpoint how rapid and delayed splicing tune the levels of alternative mRNA isoforms, ultimately altering the cellular proteome in response to developmental cues, environmental stresses, disease states, and other physiological scenarios where transcription and/or splicing can be influenced by cell signaling to achieve gene output tuned to the needs of the cell, tissue, or organism.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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